

# Benchmark summary

| model            | tests | baseline accuracy (%) | pipeline accuracy (%) | baseline time (s) | pipeline time (s) | total time (s) |
|------------------|-------|-----------------------|-----------------------|-------------------|-------------------|----------------|
| cogito:8B        | 1000  | 64.50                 | 65.60                 | 6687.71           | 7529.27           | 14216.98       |
| gemma4:e4B       | 1000  | 70.60                 | 72.70                 | 4613.96           | 9505.83           | 14119.79       |
| qwen2.5-coder:7B | 1000  | 68.90                 | 70.60                 | 4259.65           | 6889.26           | 11148.91       |
| qwen3.5:9B       | 1000  | 69.10                 | 70.40                 | 4762.45           | 10279.06          | 15041.51       |

| model            | baseline wrong | baseline correct | pipeline wrong | pipeline correct | both correct | pipeline wrong baseline correct | pipeline correct baseline wrong |
|------------------|----------------|------------------|----------------|------------------|--------------|---------------------------------|---------------------------------|
| cogito:8B        | 355            | 645              | 344            | 656              | 529          | 116                             | 127                             |
| gemma4:e4B       | 294            | 706              | 273            | 727              | 634          | 72                              | 93                              |
| qwen2.5-coder:7B | 311            | 689              | 294            | 706              | 605          | 84                              | 101                             |
| qwen3.5:9B       | 309            | 691              | 296            | 704              | 618          | 73                              | 86                              |